

Exercises on the geometry of linear equations.

Problem 1.1

$$x_1 w_1 + x_2 w_2 + x_3 w_3 = \underline{0}$$

$$w_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad w_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \quad w_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$$

$$x_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} + x_3 \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} = \underline{0}$$

* $\rightarrow$ Solve simultaneously.

* Otherwise notice

$$w_1 + w_3 - 2w_2 = 0$$

$$Ax = b$$

$$\begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

LO WHY

If solve and see you can get $= 0$ then vectors are dep.

~~$x = A^{-1}b$~~ A: Those vectors are dependent.
The three vectors lie in a plane.
The matrix W with those columns is not invertible.

*

Dependent $\rightarrow$ The vectors are dependent because there is a combination of the vectors that gives the zero vector.

Multiply: $\begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 3 \\ 4 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$

3×3 3×1

$$= \begin{bmatrix} -1 \\ 9 \\ 11 \end{bmatrix}$$

Problem 1.3

Q: True or False?

$\rightarrow$ A 3 by 2 matrix A times a 2 by 3 matrix B equals a 3 by 3 matrix AB .

$$A: A_{3 \times 2} \cdot B_{2 \times 3} = AB_{3 \times 3}$$

True

$$x \begin{bmatrix} 2 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$$A_{n \times m} \cdot B_{m \times d} = C_{n \times d}$$

$$\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$$



$$Ax = b$$

$$\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} x = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$$A^{-1}A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$x = A^{-1}b$$

$$A^{-1} = A \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = AI = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2+0+0+1=3 & 1+0+-2+0=-1 \\ 1+0+0-2=-1 & 0+1+0-2=-1 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ -1 & -1 \end{bmatrix}$$

$$x = A^{-1}b = \begin{bmatrix} 3 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 9-3-3+1=7 \\ -3+1-3+1=-4 \end{bmatrix} = \begin{bmatrix} 7 \\ -4 \end{bmatrix}$$

$$A^{-1} = AI = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & -2 & 0 & 1 \end{array} \right)$$

$\frac{1}{2} \textcircled{1}$

$$= \left(\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 1 & -2 & 0 & 1 \end{array} \right)$$

$\textcircled{2} - \textcircled{1} \quad R_2 \leftarrow R_2 - R_1$

$$= \left(\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & -\frac{5}{2} & -\frac{1}{2} & 1 \end{array} \right)$$

$$= \left(\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & \frac{1}{5} & -\frac{2}{5} \end{array} \right)$$

$R_1 \leftarrow R_1 - \frac{1}{2}R_2$

$$= \left(\begin{array}{cc|cc} 1 & 0 & \frac{2}{5} & \frac{1}{5} \\ 0 & 1 & \frac{1}{5} & -\frac{2}{5} \end{array} \right)$$

$$\frac{1}{2} - \frac{1}{10} = \frac{4}{10}$$

$$\frac{5}{10} - \frac{1}{10} = \frac{4}{10}$$

$$AA^{-1}$$

$$= I$$

$$\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} \frac{2}{5} & \frac{1}{5} \\ \frac{1}{5} & -\frac{2}{5} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\left(\frac{2}{5} \right) \left(\frac{5}{10} \right) = \frac{2}{10} = \frac{1}{5}$$

§

- Can I solve $Ax=b$ for every b ?
- Do the linear combinations of the columns fill 3-D space?

↳ For this A ; always is YES.

Target → Work with eighth dimensional "planes" in ninth dimensional space.

$$Ax = b$$

$$\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 12 \\ 7 \end{bmatrix}$$

• Idea of dot product.

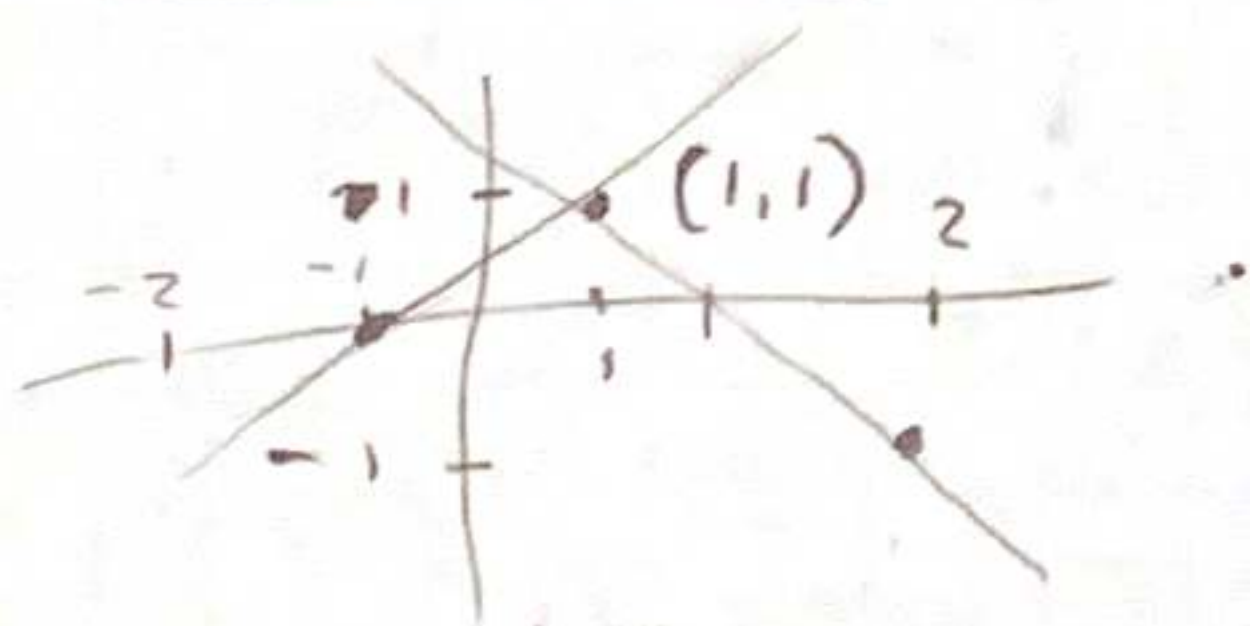
• Ax is a combination of columns.

System

$$2x + y = 3$$

$$x - 2y = -1$$

Row Picture



Matrix Form

$$Ax = b$$

$$\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = A^{-1}b$$

$$A^{-1}A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} x \\ y \end{bmatrix} = A^{-1} \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 9-3=6 \\ -3+1=-2 \end{bmatrix}$$

Col Picture

$$x \begin{bmatrix} 2 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

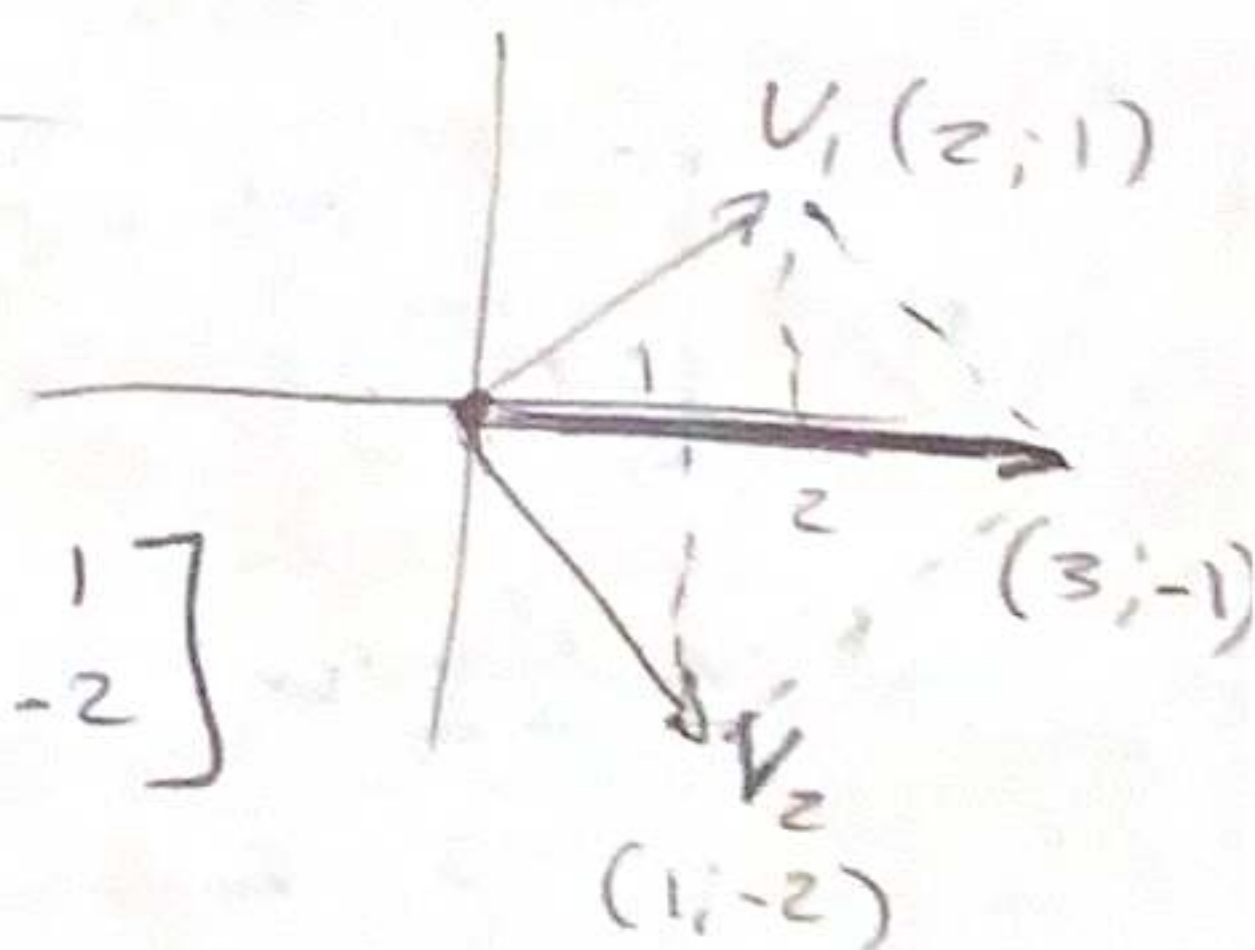
$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$$V_1 + V_2$$

$$A = [V_1 \ V_2] = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$$

=

$$ax = b \quad x = \frac{b}{a} = a^{-1}b$$



• Easy Case

n - linear equations
 n - unknowns

* Column picture $\rightarrow$

$$2x - y = 0$$

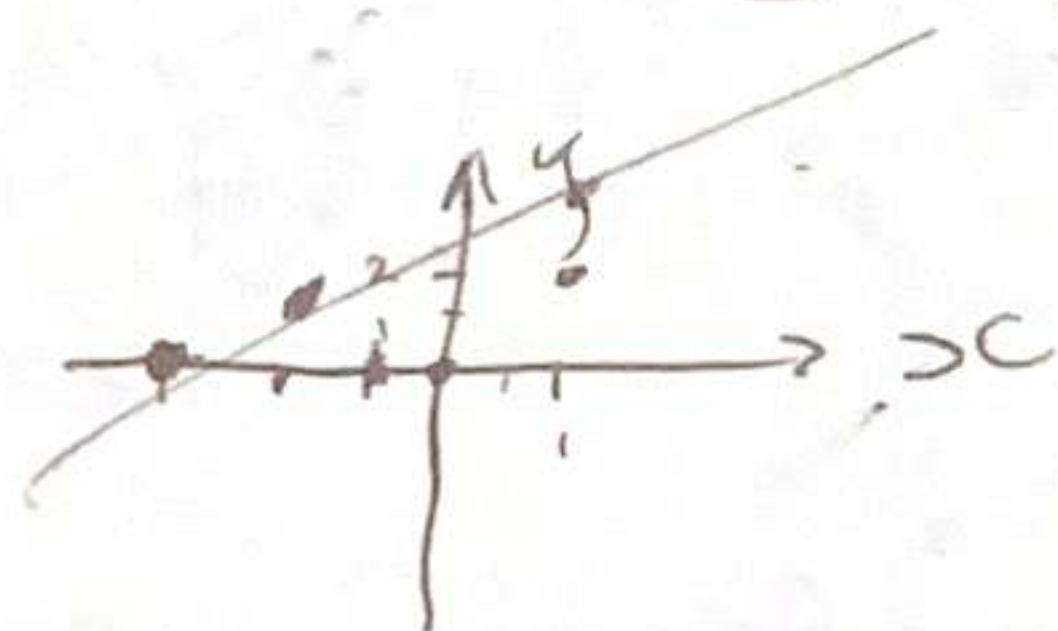
$$-x + 2y = 3$$

$$\begin{aligned} 2x &= y \\ \frac{1}{2}x + \frac{3}{2}y &= y \end{aligned}$$

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$A \quad x = b$$

• Row Picture

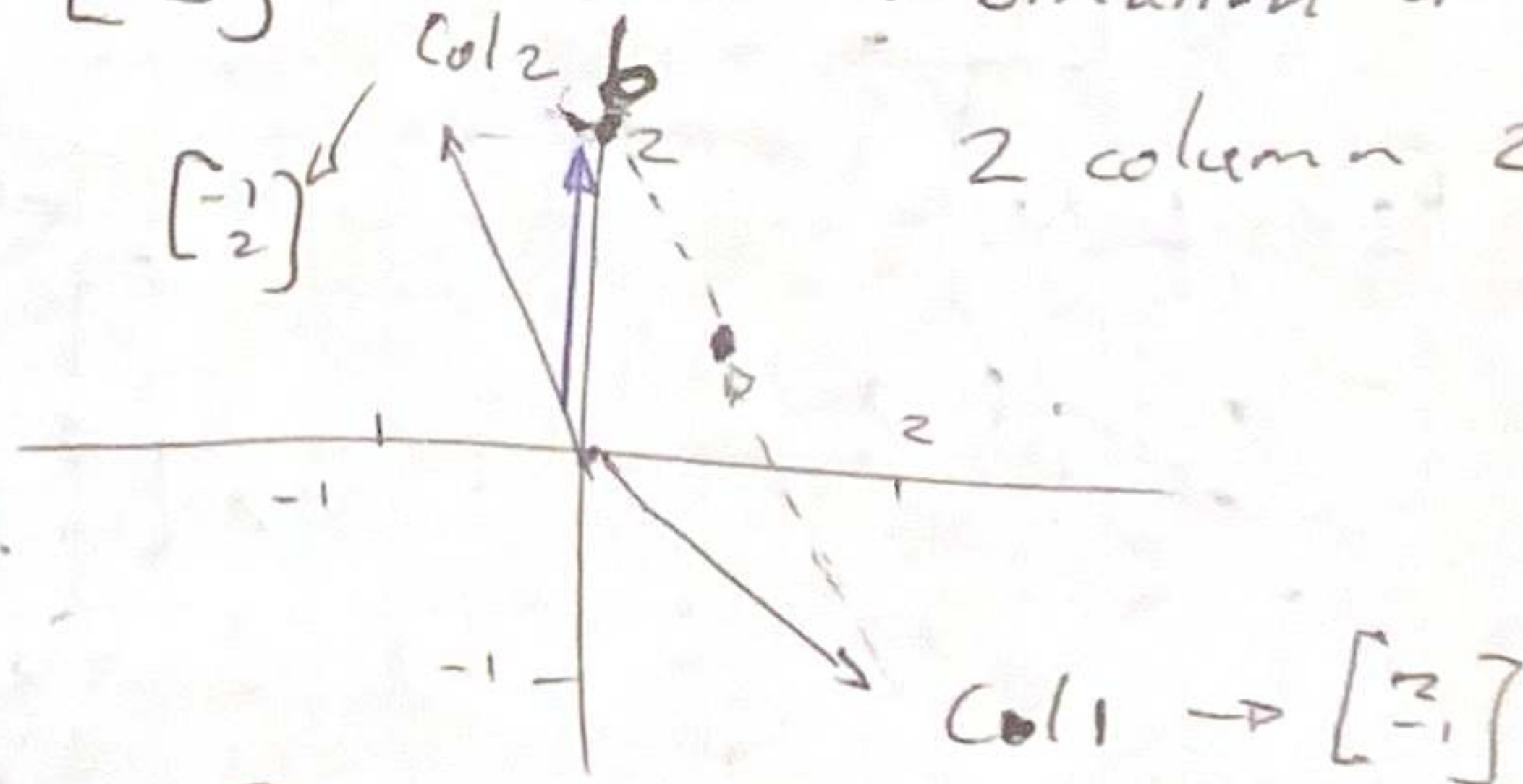


$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$x \begin{bmatrix} 2 \\ -1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$\rightarrow$ Linear combination of columns

2 columns 2

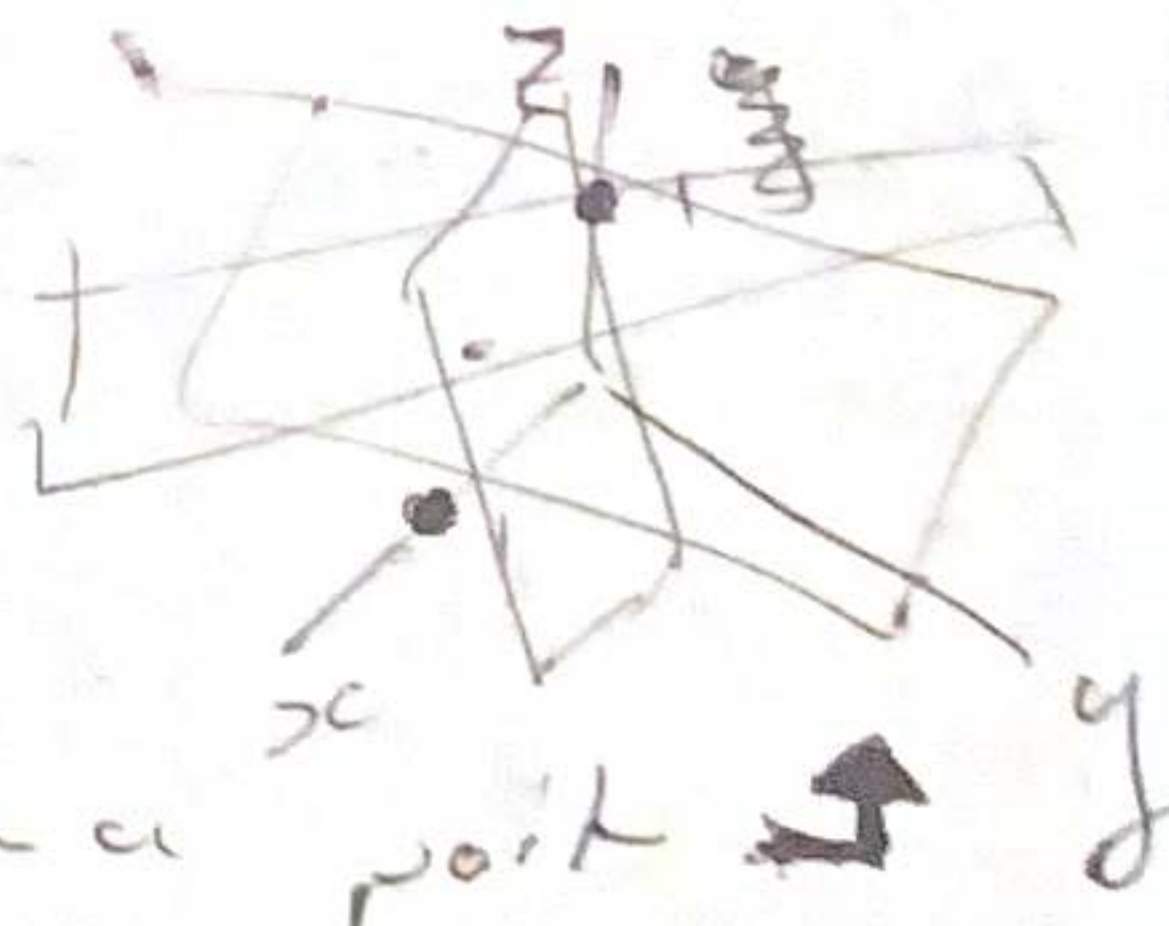


$$\begin{aligned} 2x - y &= 0 \\ -x + 2y - z &= -1 \\ -3y + 4z &= 4 \end{aligned}$$

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -3 & 4 \end{bmatrix} \quad b = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} \quad x = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Row Picture

3 equations $\Rightarrow$ 3 vars
 $\rightarrow$ 3 planes which meet in a point



$$x \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + z \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ -3 \\ -3 \end{bmatrix}$$

$$x=0 \quad y=0 \quad z=1 \quad \rightarrow \quad \begin{aligned} x &= 1 \\ y &= 1 \\ z &= 0 \end{aligned}$$

Can I solve $Ax = b$ for every b ?

• Lecture 2 - Matrix Elimination Problem Set

• Exercises on elimination with matrices

• Problem 2.1

$$\begin{array}{rcl} 2x + 3y & = & 5 \quad (1) \\ 6x + 15y & = & 12 \quad (2) \end{array}$$

Operation 1: $(2) - 3(1)$

$$\left[\begin{array}{cc|c} 2 & 3 & 5 \\ 6 & 15 & 12 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 2 & 3 & 5 \\ 0 & 6 & -3 \end{array} \right]$$

$$2x + 3y = 5$$

$$6y = -3$$

$$\therefore y = -\frac{1}{2}$$

$$2x = \frac{13}{2}$$

$$\therefore x = \frac{13}{4}$$

• Problem 2.2

$$= E \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

$$I = \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

$$A = \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{array} \right]$$

$$\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

$$= E_1 \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right] A$$

$$E_1 = \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{array} \right]$$

$$\Rightarrow E_1 A = \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \end{array} \right]$$

$$= E_2 \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right] E_1 A$$

$$E_2 = \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{array} \right]$$

$$E_2 E_1 A = \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{array} \right]$$

$$E_3 = \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 \end{array} \right]$$

$$E_3 E_2 E_1 A = \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 \end{array} \right]$$

$$\begin{aligned} M &= E_3 E_2 E_1 \\ &= E_3 \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \end{array} \right] \end{aligned}$$

• Linear Algebra - Lecture 2

• Section 2.2 Intro to Linear Algebra Textbook

• Worked Examples

2.2A) First pivot $\Rightarrow a_{11} = 1$ ✓
 Second pivot $\Rightarrow a_{22} = 1$ ✓

Multiplier $L_{21} \Rightarrow L_{21} = 1$ ✓

Pivot which would force row exchange? $\Rightarrow 0$

Why $L_{31} = 0$? a_{31} is already equal to 0 and should not be changed. ✓

A zero at the start of the row does not require elimination on

Why would Elimination fail if $a_{33} = 1$ and not 2?

A: When Row ② is subtracted from Row ③ the result will be $a_{33} = 0$ leaving a zero in the pivot position. Elimination cannot succeed with a zero in the pivot position. ✓

It implies:

$$0z = b$$

which either means there is too many solutions

$$0z = 0$$

or then, equation method request we divide by zero which we cannot do.

No third pivot
 or # eqns = 3
 pivots = 3

∴ Cannot solve

$$z = \frac{b}{0} \quad !$$

2.2B) ~~Diagonal~~ Diagonal is the pivots $\sum_{i=1}^n a_{ii}$ ✓
 Pivot along main diagonal.

One solution when? • When # equations = # pivots.

Forward Substitution $\rightarrow$ lower triangular matrix
 Backward Substitution $\rightarrow$ upper triangular matrix

2.2C)

2.2C)

$$\begin{aligned} x + y + z &= 7 \\ x + y - z &= 5 \\ x - y + z &= 3 \end{aligned}$$

(1)
(3)
(2)

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 7 \\ 1 & 1 & -1 & 5 \\ 1 & -1 & 1 & 3 \end{array} \right] = A \quad \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$$

$$\begin{aligned} \therefore x + y + z &= 7 \\ -2y &= -4 \\ -2z &= 4 \end{aligned}$$

$$\begin{aligned} (2) - (1) & \quad E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \\ (3) - (1) & \quad E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 2 & -2 \end{bmatrix} \end{aligned}$$

$$\therefore x + y + z = 7$$

$$(2) - (1): -2y = -4$$

$$\begin{aligned} (3) - (1): \\ (3) - (2): \end{aligned}$$

$$\begin{aligned} (3) + (2) & \quad E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \\ E_{32}E_{31}A &= \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} = U \end{aligned}$$

$$\begin{aligned} (5) - 7 &= -2 \\ (-2) + 4 &= 2 \end{aligned}$$

• 18.06 Linear Algebra Problem Set 2.2

1) $l_{21} = 5 \checkmark$

$$A = \begin{bmatrix} 2 & 3 \\ 10 & 9 \end{bmatrix} \quad \begin{matrix} || \\ || \end{matrix}$$

Operation

$(2) - 5(1) : \begin{bmatrix} 1 & 0 \\ -5 & 1 \end{bmatrix} = E \quad \therefore U = EA = \begin{bmatrix} 2 & 3 \\ 0 & -6 \end{bmatrix} \quad \text{pivots} = \{2, -6\}$

2) $\therefore \begin{matrix} 2x + 3y = 1 \\ -6y = 6 \\ \therefore y = -1 \\ \therefore x = 2 \end{matrix} \checkmark$

$2x + 3y = 1$
 $2(2) + 3(-1) = 4 - 3 = 1 \checkmark$

$10x + 9y = 11$
 $10(2) + 9(-1) = 20 - 9 = 11 \checkmark$

New comparison:

$$\begin{matrix} 2x + 3y = 4 \\ 10x + 9y = 44 \end{matrix}$$

$U = \begin{bmatrix} 2 & 3 \\ 0 & -6 \end{bmatrix}$
 with $(2) - 5(1)$

$$\begin{matrix} 2x + 3y = 4 \\ -6y = 24 \end{matrix}$$

$\therefore y = -4$
 $\therefore 2x + 3(-4) = 4$
 $2x = 16$
 $x = 8$

$\therefore (x, y) = (8, -4) \checkmark$

3) $\begin{matrix} 2x - 4y = 6 & (1) \\ -x + 5y = 0 & (2) \end{matrix}$
 $l_{21} = +\frac{1}{2}$

$A = \begin{bmatrix} 2 & -4 \\ -1 & 5 \end{bmatrix} \quad E =$

$\therefore U = EA = \begin{bmatrix} 2 & -4 \\ 0 & 3 \end{bmatrix} \quad \text{pivots} = \{2, 3\}$

Operation: $(2) + \frac{1}{2}(1)$

$E = \text{swap} \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix}$

$2x - 4y = 6$

$3y = 3$

Right side change ~~(1, 2)~~ $(1, 2) = (-6, 0) \quad \therefore y = 1 \quad \therefore (x, y) = (1, 1)$
 $2x - 4y = -6 \quad y = -1 \quad \therefore x = -5 \quad (x, y) = (-5, -1) \checkmark$
 $3y = -3$

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11) System of linear equations cannot have exactly two solutions?

a) Solution 1: (x, y, z)
 2: (x, y, z)
 3: $\frac{1}{2}(x+x, y+y, z+z)$

Linear systems have either

→ 1
 → 0
 → infinitely many solutions

b) If 2 planes meet at two points, they meet along the whole line through those two points.

↳ A single eqn in 3D defines a plane.
 ↳ Two points satisfy all 2 eqns lie on all 2 planes.

12) $2x + 3y + z = 8$ ①
 $4x + 7y + 5z = 20$ ②
 $-2y + 2z = 0$ ③

↳ Only geometric way two points can both lie on same plane is if the entire line through them also lies on that plane.

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & 7 & 5 \\ 0 & -2 & 2 \end{bmatrix} \quad E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \therefore \begin{bmatrix} 2 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & -2 & 2 \end{bmatrix}$$

$$\textcircled{2} - 2 \textcircled{1}$$

3

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix} \quad \therefore \begin{bmatrix} 2 & 3 & 1 \\ 0 & 1 & 3 \\ -0 & 0 & 8 \end{bmatrix}$$

$$\textcircled{3} + 2 \textcircled{2}$$

$$U = E_{32} E_{21} A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 8 \end{bmatrix}$$

$$\begin{aligned} 2x + 3y + 1z &= 8 \\ 1y + 3z &= 4 \\ 8z &= 8 \end{aligned}$$

$$\begin{aligned} \therefore z &= 1 \\ \therefore y &= 1 \\ \therefore x &= 2 \end{aligned} \quad (2, 1, 1) \checkmark$$

- Linear Algebra - Lecture 2 - Matrix Elimination Method
- Continue Problem Set 2.2 pg 53.

$$13) \begin{cases} 2x - 3y = 3 & (1) \\ 4x - 5y + z = 7 & (2) \\ 2x - y - 3z = 5 & (3) \end{cases}$$

$$A = \begin{bmatrix} 2 & -3 & 0 \\ 4 & -5 & 1 \\ 2 & -1 & -3 \end{bmatrix} \quad \begin{matrix} (2) - 2(1) \\ (3) - (1) \end{matrix}$$

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad EA = \begin{bmatrix} 2 & -3 & 0 \\ 0 & 1 & 1 \\ 2 & -1 & -3 \end{bmatrix}$$

Subtract 2 times Row 1 from Row 3 ✓

$$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & -1 \end{bmatrix} \quad E_{31}EA = \begin{bmatrix} 2 & -3 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & -3 \end{bmatrix}$$

Subtract 1 times Row 1 from Row 3 ✓

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \quad E_{32}E_{31}E_{21}A = \begin{bmatrix} 2 & -3 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & -5 \end{bmatrix}$$

Subtract 2 times Row 2 from Row 3 ✓

$$\begin{aligned} \therefore 2x - 3y &= 3 \\ \therefore 1y + 1z &= 1 \\ \therefore -5z &= 0 \end{aligned}$$

$$\begin{aligned} \therefore z &= 0 \\ \therefore y &= 1 \\ \therefore x &= 1.5 \end{aligned}$$

$$\begin{aligned} z &= 0 \\ y &= 1 \\ x &= 3 \end{aligned}$$

$$14) \begin{cases} 2x + 5y + z = 0 \\ 4x + dy + z = 2 \\ y - z = 3 \end{cases} \quad \begin{bmatrix} 2 & 5 & 1 \\ 4 & d & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

Row Exchange forced if: $d = 10$? ✓
 $d = 0$?

↳ Subtract 2 times row 1 from Row 2 to reach

$$(d-10)y - z = 2 \quad (2)$$

$d = 10 \rightarrow$ Rows 2 & 3 exchange.

If $d = 11 \rightarrow$ System becomes singular.

15)
$$\begin{aligned} x + by &= 0 \\ x - 2y + z &= 0 \\ y + z &= 0 \end{aligned}$$

$$A = \begin{bmatrix} 1 & b & 0 \\ 1 & -2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Subtract 1 times Row 1 from Row 2

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_{21}A = \begin{bmatrix} 1 & b & 0 \\ 0 & -2-b & -1 \\ 0 & 1 & 1 \end{bmatrix}$$

$-2-b=0$ would create a 0 pivot
 $b=-2$ and require a row exchange.

assume $b=-2$ $E_{21}A = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}$ $\therefore$ Exchange Row 2 with Row 3

$$\therefore \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\begin{cases} x - 2y = 0 \\ y + z = 0 \\ -z = 0 \end{cases} \Rightarrow \begin{cases} z = 0 \\ y = 0 \\ x = 0 \end{cases}$$

$$E_{21}A = \begin{bmatrix} 1 & b & 0 \\ 0 & -2-b & -1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\begin{aligned} 1 - (-2-b) &= 0 \\ 1 + 2 + b &= 0 \\ b &= -3 \end{aligned}$$

$$\textcircled{3} - \left(\frac{1}{-2-b}\right)\textcircled{2}$$

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{1}{-2-b} & 1 \end{bmatrix}$$

$$E_{32}E_{21}A = \begin{bmatrix} 1 & b & 0 \\ 0 & -2-b & -1 \\ 0 & 0 & 1 + \frac{1}{-2-b} \end{bmatrix}$$

$$\therefore 1 + \frac{1}{-2-b} = 0 \quad \{\text{Singular case}\}$$

$$\begin{aligned} \frac{1}{-2-b} &= -1 \\ 1 &= -2-b \\ b &= -3 \end{aligned}$$

$$U = \begin{bmatrix} 1 & -1 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

let $z = 1$

$$\begin{aligned} x - y &= 0 \\ -y - z &= 0 \end{aligned}$$

$$\therefore y = -z = -(+1) = -1$$

$$\therefore x = -1$$

$$(x, y, z) = (-1, -1, 1)$$

16) a) 3×3 matrix requiring 2 Row Exchanges.

$$\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\begin{aligned} 3y + 4z &= 12 \\ 3z &= 6 \\ x + y + z &= 7 \end{aligned}$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 2 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\begin{aligned} y + z &= 2 \\ z &= 3 \\ x + y + z &= 4 \end{aligned}$$

b) $\begin{bmatrix} 0 & 1 & 1 \\ 2 & 4 & 3 \\ 2 & 1 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 & 3 \\ 0 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix}$

$$2 - 3 = -1$$

17) $\begin{aligned} 2x - y + z &= 0 \\ 2x - y + z &= 0 \\ 4x - y + z &= 2 \end{aligned}$

$$A = \begin{bmatrix} 2 & -1 & 1 \\ 2 & -1 & 1 \\ 4 & 2 & 1 \end{bmatrix} \rightarrow \textcircled{2} - \textcircled{1} \quad E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_{21}A = \begin{bmatrix} 2 & -1 & 0 \\ 0 & 0 & 0 \\ 4 & 2 & 1 \end{bmatrix}$$

$\therefore$ How far $\rightarrow$ 1 elimination $\rightarrow$ 0 row $\rightarrow$ breakdown.

$$\begin{aligned} 2x + 2y + z &= 0 \\ 4x + 4y + z &= 0 \\ 6x + 6y + z &= 2 \end{aligned}$$

$$A = \begin{pmatrix} 2 & 2 & 1 \\ 4 & 4 & 1 \\ 6 & 6 & 1 \end{pmatrix} \quad E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_{31}A = \begin{bmatrix} 2 & 2 & 1 \\ 0 & 0 & -1 \\ 6 & 6 & 1 \end{bmatrix}$$

$$E_{32} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{pmatrix} \quad E_{32}E_{21}A = \begin{pmatrix} 2 & 2 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & -2 \end{pmatrix}$$

$\therefore$ Row 2 has no pivot.

18) $\begin{aligned} 1x + 2y + 4z &= b_1 \\ 6x + 12y + 24z &= b_2 \\ 5x + 10y + 20z &= b_3 \end{aligned} \quad = \begin{bmatrix} 1 \\ 6 \\ 5 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\begin{pmatrix} 1 & 2 & 4 \\ 6 & 12 & 24 \\ 5 & 10 & 20 \end{pmatrix}$$

If b_1 then $\textcircled{2} \quad 0x + 0y + 0z = 1$

$\textcircled{3} \quad 0x + 0y + 0z = 10$

$\therefore$ Infinite $\times$ No solution $Oa = k$

If b_2 then $\textcircled{2} \quad 0x + 0y + 0z = 0$

$\textcircled{3} \quad 0x + 0y + 0z = 0$

$\therefore$ Infinite

$$19) \begin{cases} x + 4y - 2z = 1 \\ x + 7y - 6z = 6 \\ 3y + qz = t \end{cases}$$

$$A = \begin{pmatrix} 1 & 4 & -2 \\ 1 & 7 & -6 \\ 0 & 3 & q \end{pmatrix} \quad b = \begin{pmatrix} 1 \\ 6 \\ t \end{pmatrix}$$

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_{21}A = \begin{bmatrix} 1 & 4 & -2 \\ 0 & 3 & -4 \\ 0 & 3 & q \end{bmatrix}$$

$$q - (-4) = 0 \Rightarrow \text{Singular}$$

$$q + 4 = 0$$

$$q = -4 \quad \checkmark$$

$$0z = t$$

$t = 0 \Rightarrow$ Infinitely many solutions

$$\text{Row 2: } 3y - 4z = 5$$

$$0y + (q+4)z = t-5 \Rightarrow 4z = t-5$$

$$((-4)+4)z = t-5$$

$$0z = t-5$$

$$t-5 = 0$$

$$t = 5$$

(achieve $0z = 0$)

$$E_2A = \begin{bmatrix} 1 & 4 & -2 \\ 0 & 3 & -4 \\ 0 & 3 & q \end{bmatrix} \quad E_2b = \begin{pmatrix} 1 \\ 5 \\ t \end{pmatrix}$$

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \quad E_{32}E_2A = \begin{bmatrix} 1 & 4 & -2 \\ 0 & 3 & -4 \\ 0 & 0 & q+4 \end{bmatrix} \quad E_{32}E_2b = \begin{bmatrix} 1 \\ 5 \\ t-5 \end{bmatrix}$$

$$\begin{aligned} (q+4)z &= t-5 & \textcircled{3} & \quad z=1 \\ 3y-4z &= 5 & \textcircled{2} & \quad y=3 \end{aligned}$$

$$\begin{aligned} z &= 1 \\ y &= 3 \end{aligned}$$

$$x + 4y - 2z = 1 \quad \textcircled{1}$$

$$x + 4(3) - 2(1) = 1$$

$$x = -9$$

20) The system is singular if row 3 of A is a combination of the first two rows.

$$x + y + z = 0$$

$$x - 2y - z = 1$$

$$2x - 1y + 0z = 1 \quad x \rightarrow 4$$

$\hookrightarrow$ anything not $\textcircled{1} \rightarrow \textcircled{2} = \text{combined}$

$\hookrightarrow$ Combine of Row 1 and Row 2.

$$\textcircled{1} + \textcircled{2} = \textcircled{3}$$

$$\begin{aligned}
 21) \quad & 2x + y = 0 \\
 & x + 2y + z = 0 \\
 & y + 2z + t = 0 \\
 & z + 2t = 5
 \end{aligned}$$

$$A = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \quad b = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 5 \end{pmatrix}$$

② - 1/2 ①

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1/2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$E_{21}A = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 3/2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

② - 1/2 ①

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -3/2 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad E_{32}E_{21}A = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 3/2 & 1 & 0 \\ 0 & 0 & 4/3 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

③ - 3/2 ②

$$E_{43} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -3/4 & 1 \end{bmatrix} \quad E_{43}E_{32}E_{21}A = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 3/2 & 1 & 0 \\ 0 & 0 & 4/3 & 1 \\ 0 & 0 & 0 & 5/4 \end{bmatrix} = U$$

$$\begin{aligned}
 2x + y + z &= 0 \\
 3/2 y + z &= 0 \\
 4/3 z + t &= 0 \\
 5/4 t &= 5
 \end{aligned}$$

$$\begin{aligned}
 \therefore t &= 4 \\
 \therefore z &= -3 \\
 \therefore y &= 2 \\
 \therefore x &= 1/2 x - 1
 \end{aligned}$$

$$\begin{aligned}
 2x + 2 &= 0 \\
 x &= -1
 \end{aligned}$$

$$\begin{aligned}
 2x - y + z &= 0 \\
 -x + 2y - z &= 0 \\
 -y + 2z - t &= 0 \\
 -z + 2t &= 5
 \end{aligned}$$

$$A = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 2 \end{bmatrix} \quad b = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 5 \end{pmatrix} \quad E_{21} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad E_{21}A = \begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & 3/2 & -1 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 3/2 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad E_{32}E_{21}A = \begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & 3/2 & -1 & 0 \\ 0 & 0 & 4/3 & 0 \\ 0 & 0 & -1 & 2 \end{bmatrix} \quad E_{43} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 3/4 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & 3/2 & -1 & 0 \\ 0 & 0 & 4/3 & 0 \\ 0 & 0 & 0 & 5/4 \end{bmatrix}$$

2 - 3/4

$$\begin{aligned}
 2x - y &= 0 \\
 3/2 y - z &= 0 \\
 4/3 z - t &= 0 \\
 5/4 t &= 5
 \end{aligned}$$

$$\begin{aligned}
 \therefore t &= 4 \\
 \therefore z &= 3 \\
 \therefore y &= 2 \\
 \therefore x &= 1
 \end{aligned}$$

} ✓

$$22) p_5 = \frac{6}{5} \quad \checkmark$$

$$p_n = \frac{n+1}{n} \quad \checkmark$$

$$23) x + y = 1$$

$$2y = 3$$

$$y = 3/2$$

$$x = -1/2$$

$$U = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \quad Eb = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$A_1 = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix} \quad A_2 = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \quad A_3 = \begin{bmatrix} 1 & 1 \\ 3 & 5 \end{bmatrix}$$

$$2y = 3$$

$$2y + l(x+y) = \cancel{3} + l$$

$$2y + 2(x+y) = 3 + 2$$

$$2x + 4y = 5$$

$$2(-1/2) + 4(3/2) =$$

$$= -1 + 6$$

$$= 5$$

$$24) A = \begin{bmatrix} a & 2 \\ a & a \end{bmatrix}$$

$$a = 2 \rightarrow \text{Row 2} = 0 \quad \checkmark$$

$$a = 0 \rightarrow \text{Row 2} = 0 \quad \checkmark$$

$$25) A = \begin{bmatrix} a & 2 & 3 \\ a & a & 4 \\ a & a & a \end{bmatrix}$$

$$a = 0 \quad \checkmark$$

$$a = 2 \quad \checkmark$$

$$a = 4 \quad \checkmark$$

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• Problem Set 2.4

$$A_{m \times n} \times B_{n \times p} = C_{m \times p}$$

1) $A_{3 \times 5}$ $B_{5 \times 3}$ $C_{5 \times 1}$ $D_{3 \times 1}$

$BA \Rightarrow$ Allowed ✓

$AB \Rightarrow$ ~~Not~~ Allowed ✓

$ABD \Rightarrow AB_{3 \times 3} \Rightarrow$ Allowed ✓

$DC \Rightarrow$ Not allowed ✓

$A(B+C) \Rightarrow$ Allowed ✓

$BA_{3 \times 3}$

$AB_{5 \times 5}$

$ABD_{3 \times 1}$

$\rightarrow$ Not defined

$A(B+C)_{3 \times 3} \rightarrow$ Not defined

~~$A(B+C)$~~

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

5) $A = \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix} \therefore A^2 = \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2b \\ 0 & 1 \end{bmatrix}$

$\therefore A^3 = \begin{bmatrix} 1 & 2b \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3b \\ 0 & 1 \end{bmatrix}$

$\therefore A^5 = \begin{bmatrix} 1 & 5b \\ 0 & 1 \end{bmatrix}$

$A^n = \begin{bmatrix} 1 & nb \\ 0 & 1 \end{bmatrix}$

$A = \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} \therefore A^2 = \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 0 & 0 \end{bmatrix}$

$\therefore A^3 = \begin{bmatrix} 4 & 4 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 8 & 8 \\ 0 & 0 \end{bmatrix}$

$\therefore A^5 = \begin{bmatrix} 32 & 32 \\ 0 & 0 \end{bmatrix}$

$A^n = \begin{bmatrix} 2^n & 2^n \\ 0 & 0 \end{bmatrix}$

7) a) True

$$A^2 = \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 0 & 0 \end{bmatrix}$$

b) True

x

$$\begin{bmatrix} 1 & 3 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}$$

False

False

True if Rows
A are same

$$\begin{bmatrix} 1 & 3 & 1 \\ 2 & 1 & 2 \\ 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 5 & 6 \\ 1 & 2 & 1 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 12 & 10 \\ 11 & 14 & 15 \\ 8 & 12 & 10 \end{bmatrix}$$

c) True

d) True

x

$$(AB)^2$$

False

$$(AB)^2 = ABAB \neq A^2B^2$$

9) a) $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $F = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ $E = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

$$AF = \begin{bmatrix} a & a+b \\ c & c+d \end{bmatrix}$$

$$E(AF) = \begin{bmatrix} a & a+b \\ a+c & a+b+c+d \end{bmatrix}$$

b) $(EA)F = E(AF)$ $\therefore$ Associative law of Matrix Multiplication holds

11) $A = \begin{bmatrix} 2 & 3 \\ 5 & 9 \end{bmatrix}$ Row Operation: $\textcircled{1} \times 2$ $\begin{bmatrix} 4 & 6 \\ 5 & 9 \end{bmatrix}$

Col Operation: $\textcircled{2} \div 3$ $\begin{bmatrix} 4 & 2 \\ 5 & 3 \end{bmatrix}$

Reverse Operation Order:

Col Operation: $\textcircled{2} \div 3$ $\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$

Row Operation: $\textcircled{1} \times 2$ $\begin{bmatrix} 4 & 2 \\ 5 & 3 \end{bmatrix}$

~~12)~~

15) a) True

b) ~~True~~ False

$$\begin{matrix} A_{2 \times 3} & B_{2 \times 3} \\ A_{3 \times 2} & B_{2 \times 3} \end{matrix}$$

$$A_{2 \times 3}$$

$$A_{2 \times 3}$$

$$A_{3 \times 2}$$

$$B_{2 \times 3}$$

c) True

d) True x False

$$(b=0)$$

$$A(0) = 0$$

~~to A~~
 $\therefore A$ could be anything

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Problem Set 2.5

1) $A = \begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix}$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix}^{-1} = \frac{1}{(0)(0)-(3)(4)} \begin{bmatrix} 0 & -3 \\ -4 & 0 \end{bmatrix}$$

$$= -\frac{1}{12} \begin{bmatrix} 0 & -3 \\ -4 & 0 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 0 & -\frac{1}{4} \\ \frac{1}{3} & 0 \end{bmatrix} \checkmark$$

$$B^{-1} = \begin{bmatrix} 2 & 0 \\ 4 & 2 \end{bmatrix}^{-1} = \frac{1}{(2)(2)-(0)(4)} \begin{bmatrix} 2 & 0 \\ -4 & 2 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 2 & 0 \\ -4 & 2 \end{bmatrix}$$

$$B^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ -1 & \frac{1}{2} \end{bmatrix} \checkmark$$

$$[A \ I] = \left[\begin{array}{cc|cc} 3 & 4 & 1 & 0 \\ 5 & 7 & 0 & 1 \end{array} \right] \Rightarrow A^{-1} = \frac{1}{(3)(7)-(4)(5)} \begin{bmatrix} 7 & -4 \\ -5 & 3 \end{bmatrix}$$

$$\Rightarrow \therefore [I \ A^{-1}] \Rightarrow \left[\begin{array}{cc|cc} 1 & -\frac{1}{3} & \frac{5}{3} & 0 \\ 5 & 7 & 0 & 1 \end{array} \right] = \frac{1}{1} \begin{bmatrix} 7 & -4 \\ -5 & 3 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 7 & -4 \\ -5 & 3 \end{bmatrix} \checkmark$$

3) $\begin{bmatrix} 10 & 20 \\ 20 & 50 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$10x + 20y = 1$$

$$20x + 50y = 0$$

$$2x = -5/2 y$$

$$10(-5/2 y) + 20y = 1$$

$$-25y + 20y = 1$$

$$y = -1/5$$

$$\therefore x = 1/2$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1/2 \\ -1/5 \end{bmatrix} \checkmark$$

$\begin{bmatrix} 10 & 20 \\ 20 & 50 \end{bmatrix} \begin{bmatrix} t \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$10t + 20z = 0 \quad (1) \quad t = -2z$$

$$20t + 50z = 1 \quad (2)$$

$$\therefore 20(-2z) + 50z = 1$$

$$10z = 1$$

$$z = 1/10$$

$$\therefore z = 1/10$$

$$\therefore t = -1/5$$

$$\begin{bmatrix} t \\ z \end{bmatrix} = \begin{bmatrix} -1/5 \\ 1/10 \end{bmatrix} \checkmark$$

5) $u = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$7) \text{ Row } 1 + \text{Row } 2 = \text{Row } 3$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 5 & 7 & 9 \end{bmatrix} \begin{bmatrix} a & b & c \end{bmatrix} = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}$$

$y, z \quad x, y, z$

$$\text{eqn } 1 + \text{eqn } 2 - \text{eqn } 3 = 0$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 5 & 7 & 9 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$10) A = \begin{bmatrix} 0 & 0 & 0 & 2 \\ 0 & 0 & 3 & 0 \\ 0 & 4 & 0 & 0 \\ 5 & 0 & 0 & 0 \end{bmatrix} \quad [A \quad I] \Rightarrow \left[\begin{array}{cccc|cccc} 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & 0 & 3 & 0 & 0 & 0 & 1 & 0 \\ 0 & 4 & 0 & 0 & 0 & 1 & 0 & 0 \\ 5 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{array} \right]$$

$$\Rightarrow \begin{bmatrix} 0 \\ 0 \\ 0 \\ 5 \end{bmatrix}$$

• Linear Algebra - Lecture 3 - Problem Set

• Matrix Multiplication & Inverse Matrices

• Problem 1.3

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix} \quad AC = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} 10 & 12 \\ 20 & 24 \end{bmatrix} \quad AB+AC = \begin{bmatrix} 11 & 12 \\ 23 & 24 \end{bmatrix}$$

$$A(B+C) = A \left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 5 & 6 \end{bmatrix} \right) = A \begin{bmatrix} 1 & 0 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} 11 & 12 \\ 23 & 24 \end{bmatrix}$$

$$\therefore AB+AC = A(B+C) \quad \checkmark$$

• Problem 3.2

$$UU^{-1} = I \quad \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \quad I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[U \ I] = \left[\begin{array}{ccc|ccc} 1 & a & b & 1 & 0 & 0 \\ 0 & 1 & c & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] = \left[\begin{array}{ccc|ccc} 1 & 0 & b-ac & 1 & -a & 0 \\ 0 & 1 & c & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & b-ac & 1 & -a & 0 \\ 0 & 1 & 0 & 0 & 1 & -c \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \quad \begin{array}{l} (b-ac) - (+b-ac)(1) \\ b-ac - b+ac \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & -a & ac-b \\ 0 & 1 & 0 & 0 & 1 & -c \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] = [I \ U^{-1}]$$

$$U^{-1} = \begin{bmatrix} 1 & -a & ac-b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{bmatrix} \quad \checkmark$$